1. Core difference between Case Study A and B

Case Study A focuses on **schema alignment and data transformation**. The pipeline reads structured sales data and converts it into a consistent format for analysis.

Case Study B focuses more on **data validation and structured inspection**. It handles real-world healthcare data where cleaning, validation, and handling inconsistencies are important before generating reports.

2. What role does the LLM play?

The LLM does not directly modify files. Instead, it **analyzes the data and generates instructions or structured outputs**, which are then used by the system to process and transform the data.

In simple terms, the LLM acts like a **decision-maker or guide**, not the one doing the actual file changes.

3. What happens if:

- The API key is missing?

  The pipeline will fail to connect to the LLM, and AI-based processing steps will not work. An error will be generated.

- Dependencies are not installed?

  The program will fail to run, and errors like module not found will occur.

- The file format is wrong?

  The pipeline may fail during reading or processing, or produce incorrect results due to mismatched structure.

Explain the Full Flow

The pipeline starts with raw CSV files, which are processed by the AI agent to produce structured outputs. These outputs are then cleaned, validated, and transformed into a consistent format. Finally, the processed data is used to generate a report.

FLOW:

Raw CSV Files

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AI Agent (LLM Analysis)

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Structured Output / Instructions

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Data Processing (Merge + Clean + Validate)

↓

Cleaned Data

↓

Report Generation

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Final Output (CSV / PDF)

Analogy

The AI pipeline works like a **teacher checking assignments**.
The raw data is like a student's rough work.
The AI agent reviews and corrects it, and finally produces a clean and proper report ready for submission.